PROJECT DOCUMENTATION

Multimodal Detection of
Fraudulent Product Returns

Using Vision-Language Models and Claim-Consistency Scoring in E-Commerce

BUSINESS • TECHNICAL • EXECUTION

Document Control	Details
Document version	1.0
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Client / organisation	KartNova Retail Pvt. Ltd. (illustrative fictional organisation)
Document audience	Project Manager, Data Scientist / ML Intern, ML Engineer, Data Engineer, Business Reviewer
Document status	Project proposal and execution baseline; subject to stakeholder approval

CONFIDENTIALITY NOTE

This document contains a proposed solution design. The client name, business estimates, KPI baselines and targets are illustrative and must be validated against approved business data before being used for operational or financial decisions.

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1. Executive Summary

KartNova Retail's returns operation is proposed to use multimodal AI to identify potentially fraudulent or abusive product-return claims. The solution analyses customer-submitted images, complaint text, product-catalog information and transaction/customer history to support consistent, explainable decisions.

The core outputs are a Claim-Consistency Score (CCS), a fraud-risk score, supporting explanations and a recommended workflow action. The system is designed as decision support; adverse decisions remain subject to human review during the initial phase.

Illustrative business outcomes

KPI	Illustrative baseline	Target
Fraud recall at 80% precision	35%	70% or higher
False-positive rate on genuine customers	6%	Below 2%
Manual review rate	All flagged cases reviewed	Reduce by 40%
Average decision time	18 hours	Under 5 minutes for automated processing
Annual fraud-loss reduction	Not established	25–35%

2. Business Context and Problem Statement

2.1 Background

Return rates in Indian fashion, electronics and home categories range from 8 to 30 percent. Each return costs logistics, inspection, refurbishment and often a write-off. Fraud rings exploit "no questions asked" policies.

2.2 Problem Statement

Given a return request consisting of images, text and metadata, determine whether the claim is consistent with the evidence and estimate the probability that the return is fraudulent or abusive, with an explanation a human reviewer can verify.

2.3 Why Vision-Language Models

Rule-based and tabular models cannot tell whether a photo of a cracked screen actually belongs to the ordered phone model, or whether the complaint "colour is different" matches what the picture shows. VLMs embed images and text in a shared space, so semantic agreement between a claim and an image can be measured directly.

2.4 Business Objectives and KPIs

KPI Baseline Target

Fraud recall at fixed precision of 80% 35% 70%+

False positive rate on genuine customers 6% under 2%

Manual review rate 100% flagged reduce by 40%

Average decision time 18 hours under 5 min (auto)

Annual fraud loss reduction n/a 25 to 35%

2.5 Scope

In scope: image + text + metadata analysis, CCS, fraud risk scoring, explanation output, REST API, reviewer dashboard, monitoring.

Out of scope: (Phase 1) video evidence, courier-side fraud, payment fraud, warehouse inspection automation.

3. Stakeholders and Team Responsibilities

approval

Project Manager: plan, risk tracking, client communication

Data Scientist / Intern: EDA, modelling, evaluation (primary executor of this document)

ML Engineer: pipeline, API, deployment

Data Engineer: ingestion, storage, feature pipeline

Domain SME (Returns Ops): fraud pattern labelling and review

Legal / Privacy Officer: DPDP Act 2023 compliance review

4. Solution Architecture

4.1 High-Level Flow

Customer raises a return request (app/web) with photos, reason code and free text.

Ingestion service stores the request, images (object storage) and metadata (database).

IMAGE PREPROCESSING: resize, normalise, quality check (blur, brightness), duplicate / reverse-image check, EXIF extraction.

TEXT PREPROCESSING: language detection, translation to English for Hindi, Tamil and Hinglish, cleaning, intent classification.

VLM MODULE: generates image embeddings and captions, answers targeted questions about the image (VQA).

CLAIM-CONSISTENCY SCORING: compares the claim, catalog listing and image evidence.

BEHAVIOURAL FEATURE MODULE: customer and order history features.

FRAUD RISK MODEL: combines CCS and behavioural features.

DECISION ENGINE: threshold rules produce Approve, Review, Reject or Request-more-evidence.

EXPLANATION AND DASHBOARD: highlights the reason for the score for human reviewers.

FEEDBACK LOOP: reviewer decisions and confirmed outcomes are fed back for retraining.

4.2 Component Summary

Storage: S3 / Azure Blob (images), PostgreSQL (metadata)

Preprocessing: Python, OpenCV, Pillow, langdetect, IndicTrans2 or Google Translate API

VLM: CLIP / SigLIP (embeddings); BLIP-2, LLaVA or Qwen2-VL (captioning and VQA)

Scoring and ML: PyTorch, scikit-learn, XGBoost / LightGBM

Experiment tracking: MLflow

Serving: FastAPI, Docker, Kubernetes

Dashboard: Streamlit or React + Grafana

Monitoring: Evidently AI, Prometheus

5. Data Description and Governance

5.1 Data Sources (Production)

- Return requests table
- Order and catalog table (product title, category, listing images, price)
- Customer history table
- Customer-uploaded images (1 to 5 per return)
- Agent review outcomes (ground-truth labels)

5.2 Synthetic Dataset Provided

Because real data cannot be shared, a synthetic dataset generator and sample file are provided: return_claims_synthetic.csv (about 20,000 rows, around 9% fraud) and generate_synthetic_returns.py. It mimics real distributions: class imbalance, seasonal spikes (festive sale periods), serial returners and fraud-pattern signatures. Images are represented by placeholder file paths and precomputed proxy scores, because real images cannot be fabricated by a script. For image experiments use public datasets (see 5.4).

IMPORTANT: the fraud label and the proxy scores were generated together, so models will score unrealistically well. Use the file to build and test the pipeline, not as a benchmark.

5.3 Data Dictionary

return_id string Unique return request ID

order_id string Linked order

customer_id string Anonymised customer

request_date date Date of return request

category cat. Fashion, Electronics, Home, Beauty, Appliances, Footwear

product_title string Catalog title

order_value_inr float Order amount in INR

days_since_delivery int Days from delivery to return request

return_reason_code cat. Damaged, Wrong item, Not as described, Size issue, Missing parts, Quality issue

complaint_text string Free-text complaint

complaint_language cat. en, hi, ta, hinglish

num_images int Images uploaded

image_paths string Semicolon-separated image paths (placeholders)

image_blur_score float 0 (sharp) to 1 (very blurry)

exif_timestamp_gap_days int Gap between image capture time and

delivery date (negative = taken before
delivery)
image_reuse_flag int 1 if image hash seen in other claims
clip_image_text_sim float Proxy for CLIP similarity between image
and complaint text
clip_image_listing_sim float Proxy for similarity between uploaded
image and catalog image
vqa_defect_visible int 1 if VQA confirms the claimed defect
claim_consistency_score float CCS in 0 to 1 (higher = more consistent)
cust_total_orders_12m int Orders in last 12 months
cust_return_rate_12m float Return rate in last 12 months
cust_account_age_days int Account age
cust_prev_fraud_flags int Prior confirmed fraud flags
refund_requested_amount float Amount requested
payment_mode cat. COD, UPI, Card, NetBanking, Wallet
delivery_pincode_risk float Historical fraud rate of pincode (0 to 1)
is_fraud int TARGET: 1 = fraud/abusive, 0 = genuine
fraud_type cat. none, wardrobing, item_swap, empty_box,
false_damage, not_as_described_abuse

5.4 Public Datasets to Supplement (for real image work)

- DeepFashion / Fashion-MNIST style sets for apparel images
- Amazon Reviews (Berkeley) for product text and images
- Damage-detection sets from Kaggle (cracked screens, product damage)
- Flickr30k / COCO captions for VLM sanity testing

5.5 Privacy and Compliance

Anonymise customer IDs, strip EXIF GPS data, store images encrypted, apply role-based access, follow India's DPDP Act 2023 and the client's retention policy. Get Legal sign-off before using any real data.

6. Delivery Plan and Technical Execution

PHASE 0: KICK-OFF AND SETUP (Week 1)

Read this document fully and the client problem brief.

Set up the environment: Python 3.11, virtual environment, Git repository, folder structure (data/, notebooks/, src/, models/, reports/, api/).

Install libraries: torch, transformers, open_clip_torch, scikit-learn, xgboost, lightgbm, pandas, opencv-python, mlflow, fastapi, shap.

Create the project tracker (Jira / Trello) with the tasks below.

Deliverable: working environment and repo, architecture sign-off.

PHASE 1: DATA UNDERSTANDING AND EDA (Weeks 1 to 2)

Load return_claims_synthetic.csv; check shape, dtypes, missing values, duplicates.

Class balance: confirm approximately 9% fraud; discuss impact on metrics.

Univariate analysis of every column; histograms and boxplots.

Bivariate analysis against is_fraud (fraud rate by category, payment mode, return reason, days since delivery).

Temporal analysis: returns by month, festive spikes.

Customer-level analysis: serial returners, new accounts with high-value returns.

Text analysis: word clouds and top n-grams for fraud vs genuine complaints.

Correlation matrix and mutual information with the target.

Document 10 or more insights in the EDA report.

Deliverable: EDA notebook and report (reports/eda_report.md).

PHASE 2: DATA PREPROCESSING (Week 3)

Tabular

Handle missing values (median for numeric, "unknown" for categorical).

Cap outliers at the 99th percentile for amounts and days.

Encode categoricals (target encoding for pincode and product, one-hot for low-cardinality).

Scale numeric features for neural models.

Text

Detect language; translate Hindi, Tamil and Hinglish to English (or use a multilingual encoder).

Normalise text: lowercase, remove URLs and emojis, expand common abbreviations.

Keep an original-text copy for the dashboard.

Image

Validate files (corrupt, too small, wrong format).

Compute blur (variance of Laplacian), brightness and resolution flags.

Perceptual hash (pHash) for duplicate and reuse detection across customers.

Extract EXIF capture timestamp and compare with delivery date.

Resize to the VLM input size (224 or 336) and normalise.

Splitting

Use a TIME-BASED split (train: first 70% of dates, validation: next 15%, test: last 15%).

Group by customer_id so the same customer never appears across splits.

Deliverable: clean data, preprocessing pipeline (src/preprocess.py), data-quality report.

PHASE 3: VISION-LANGUAGE FEATURE EXTRACTION (Weeks 4 to 5)

EMBEDDINGS. Use CLIP or SigLIP to encode (a) uploaded images, (b) catalog listing images, (c) complaint text, (d) product title.

SIMILARITY FEATURES.

sim_img_text: cosine(uploaded image, complaint text)

sim_img_listing: cosine(uploaded image, listing image)

sim_text_title: cosine(complaint, product title)

CAPTIONING. Generate captions of uploaded images with BLIP-2 / LLaVA / Qwen2-VL.

VQA PROBING. Ask targeted questions, for example: "Is the screen cracked?", "Is the product in the image the same as a {product_title}?", "Is the box empty?", "Does the item look used or worn?" Store yes/no with confidence.

OBJECT AND DEFECT CHECK. Optionally add a defect detector (YOLOv8 fine-tuned on damage images).

Cache all embeddings to disk (Parquet / FAISS) to avoid recomputation. Deliverable: feature table vlm_features.parquet, feature extraction notebook and script.

PHASE 4: CLAIM-CONSISTENCY SCORING (Weeks 5 to 6)

The CCS quantifies whether WHAT THE CUSTOMER SAYS matches WHAT THE EVIDENCE SHOWS.

4.1 Components

C1 Text-Image agreement: does the image support the complaint? (CLIP similarity, caption entailment)

C2 Listing-Image identity: is the returned item the same as ordered? (image-to-listing similarity, fine-grained match)

C3 Defect verification: is the claimed defect visible? (VQA / defect detector)

C4 Reason-Text consistency: does the free text match the selected reason code? (zero-shot text classifier)

C5 Metadata plausibility: is the evidence plausible in time? (EXIF gap, image reuse, blur)

4.2 Baseline formula (rule-based)

CCS = w1*C1 + w2*C2 + w3*C3 + w4*C4 + w5*C5

Initial weights: 0.25, 0.25, 0.25, 0.10, 0.15 (all components scaled 0 to 1). Apply a hard penalty of 0.5 if image_reuse_flag = 1.

4.3 Learned version

Train a small MLP or logistic regression on the five components (plus raw similarities) to predict a "claim-consistent" label derived from reviewer decisions. Calibrate output with Platt scaling or isotonic regression so CCS behaves like a probability.

4.4 Validation

- Check that CCS distributions differ clearly between genuine and

fraudulent claims (KS test, AUC of CCS alone).

- Run ablations removing each component.
- Deliverable: src/ccs.py, CCS validation report.

PHASE 5: FRAUD RISK MODELLING (Weeks 6 to 7)

Baselines: logistic regression on tabular features; XGBoost on tabular features only.

Multimodal model: XGBoost / LightGBM on tabular + VLM similarities + CCS + behavioural features.

Optional deep fusion: a neural network that concatenates image embedding, text embedding and tabular vector (late fusion), trained with focal loss.

Class imbalance handling: class weights, focal loss, SMOTE only on training folds, threshold tuning (not just accuracy).

Hyperparameter tuning: Optuna with stratified grouped k-fold.

Experiment tracking: log every run in MLflow (parameters, metrics, artifacts).

Deliverable: model comparison table, best model artifact, model card.

PHASE 6: EVALUATION (Week 8)

Metrics:

PR-AUC main metric under class imbalance

ROC-AUC overall ranking quality

Recall at precision 80% operating point for the business

False positive rate (genuine) customer experience

F1 / F2 threshold selection

Expected cost saved (INR) (fraud caught x avg refund) minus (review cost + friction cost)

Also report: confusion matrix, calibration curve, performance by category, language and payment mode (fairness check), and error analysis with 20 or more misclassified examples.

Ablation table to include: tabular only, + text, + image, + CCS, + all.

PHASE 7: EXPLAINABILITY (Week 8)

SHAP values for global and per-claim explanations.

Human-readable reasons, for example: "Image does not match listing (similarity 0.31)", "Claimed crack not visible", "Image reused in 3 other claims".

Highlight image regions (Grad-CAM or attention maps) when feasible.

Deliverable: explanation module and sample explanation reports.

PHASE 8: DECISION ENGINE (Week 9)

Fraud Risk Score CCS Action

under 0.2 above 0.7 Auto-approve refund

0.2 to 0.5 any Standard review queue

0.5 to 0.8 below 0.5 Priority manual review + request extra photos

above 0.8 below 0.4 Hold refund, escalate to fraud team

Thresholds must be tuned with the client and reviewed monthly. Never auto-reject a customer on the model alone in Phase 1; always keep a human in the loop for adverse decisions.

PHASE 9: DEPLOYMENT (Weeks 9 to 10)

Wrap the pipeline in a FastAPI service: POST /v1/score-return accepts JSON with images (URLs) and text; returns CCS, risk score, action, reasons.

Containerise with Docker; push to the registry.

Deploy on Kubernetes with a GPU node pool for VLM inference (or use quantised models on CPU for the pilot).

Target latency under 3 seconds per request (p95); batch mode for backlog.

Build the reviewer dashboard (queue, image viewer, score breakdown, approve/reject buttons).

CI/CD with unit tests, integration tests and a smoke test.

Deliverable: deployed staging environment, API documentation (OpenAPI).

PHASE 10: MONITORING AND FEEDBACK LOOP (Week 11 onwards)

Monitor data drift (Evidently), prediction drift, latency and error rate.

Track weekly precision and recall using reviewer outcomes.

Capture reviewer overrides as new labels; retrain monthly or when drift alerts fire.

Watch for adversarial adaptation by fraudsters (AI-generated images). Add AI-image detection as a later enhancement.

PHASE 11: PILOT, HANDOVER AND CLOSURE (Week 12)

A/B pilot on 10% of traffic in two categories.

Present KPI results to the client sponsor.

Hand over code, documentation, model card and runbook.

Deliverable: final report and presentation.

7. Project Timeline

Week Activity Milestone

1 Setup, requirements, EDA start Kick-off complete

2 EDA EDA report

3 Preprocessing Clean data + pipeline

4 to 5 VLM features Feature store

5 to 6 CCS design and validation CCS v1

6 to 7 Fraud modelling Best model selected

8 Evaluation + explainability Evaluation report

9 Decision engine + API Staging API

10 Dashboard, testing UAT ready

11 Monitoring setup Production readiness review

12 Pilot, handover Closure

8. Deliverables and Acceptance Checklist

- ■ Project charter and this documentation
- ■ EDA report and notebook
- ■ Preprocessing pipeline code
- ■ VLM feature extraction code and feature store
- ■ CCS module and validation report
- ■ Trained models, MLflow experiment logs, model card
- ■ Evaluation and ablation report
- ■ Explainability module
- ■ API, Docker image, dashboard
- ■ Monitoring dashboard and runbook
- ■ Final presentation and handover document

9. Risks and Mitigations

Risk: Severe class imbalance -> poor recall

Mitigation: class weights, PR-AUC, threshold tuning

Risk: Noisy fraud labels -> wrong learning

Mitigation: reviewer audits, label confidence, active learning

Risk: Genuine customers wrongly flagged -> brand damage

Mitigation: human-in-loop, low FPR target, appeal flow

Risk: VLM hallucination in captions / VQA -> wrong CCS

Mitigation: multiple signals, calibration, confidence thresholds

Risk: Fraudsters use AI-generated/reused images -> model bypass

Mitigation: image hash, AI-image detector, periodic retraining

Risk: Privacy and legal exposure -> compliance issue

Mitigation: DPDP-aligned retention, anonymisation, Legal sign-off

Risk: GPU cost and latency -> budget overrun

Mitigation: quantised models, cached embeddings, batch scoring

Risk: Domain shift across categories -> accuracy drop

Mitigation: per-category evaluation and thresholds

Risk: Bias against regions or languages -> fairness issue

Mitigation: subgroup metrics, regular audits

10. Ethics and Responsible AI

- Model outputs are decision support; adverse actions require human review.
- Provide customers a way to contest decisions.
- Do not use protected attributes (religion, caste, gender) as features;

audit proxy variables such as pincode.

• Maintain audit logs of every score and decision.

11. Working Guidelines for the Project Team

Work in small, reviewed steps: raise a pull request at the end of each phase.

Never train and test on the same customers; avoid data leakage (for example, features computed using future data).

Always report PR-AUC and recall at fixed precision, not accuracy.

Keep every experiment reproducible: fixed seeds, logged configs, versioned data.

Write a short weekly status note: done, blocked, next.

Explain results in business language (money saved, customers affected) as well as in metrics.

12. Glossary

VLM Vision-Language Model that jointly understands images and text

CLIP/SigLIP Models that embed images and text in a shared space

VQA Visual Question Answering

CCS Claim-Consistency Score: consistency between claim and evidence

Wardrobing Buying, using and returning an item

PR-AUC Area under the precision-recall curve

GMV Gross Merchandise Value

13. References and Suggested Reading

• Radford et al., "Learning Transferable Visual Models From Natural Language Supervision" (CLIP)

• Li et al., "BLIP-2: Bootstrapping Language-Image Pre-training"

• Zhai et al., "Sigmoid Loss for Language Image Pre-Training" (SigLIP)

• Lin et al., "Focal Loss for Dense Object Detection"

• Lundberg and Lee, "A Unified Approach to Interpreting Model Predictions" (SHAP)

• India Digital Personal Data Protection Act, 2023

Document Governance

Version	Date	Description	Status
1.0	23 October 2026	Initial consolidated project documentation	For review

Approval checkpoints: business objectives and KPI definitions; data access and privacy review; model evaluation criteria; decision thresholds; pilot-readiness review; and final handover.

Important implementation note: all performance targets, data characteristics and business-loss estimates in this document are planning assumptions until validated with the client and measured on representative, independently labelled data.